TorqCapture — What The Event Needs

Same base as TorqFall (rented cars, turf, show layer, drivers' stand, crew, ops — see that doc; it all reuses, and unlike TorqSiege the ***floor plan also mostly reuses*** *— open turf, no obstacle gauntlet, just lit rings on it). This is the* ***delta*** *— the zone kit — plus the* ***full standalone-event number*** *with the same ₹4L envelope check TorqFall and TorqSiege get, because TorqCapture is its own event per the locked decisions. Rented stock cars → every tag/beacon is strap-on/velcro/zip-tie, nothing screwed to a rented car (same rule as the foam bump-rings).* ***The one genuinely new spend is the electronics*** *— the scoring rig + optional lit rings; everything else is reused base.* ***Prices verified against live India listings, Sept 2026*** *(sources at the bottom).* ***Electronics numbers are pulled straight from [[TorqCapture — Zone-Capture Tech]]'s costed BOM — not re-invented here; other quantities derive from the Rulebook §3–§5****: 2 teams × 2–3 cars, 3 zones (Domination) or 1 (single-zone modes), tags = fleet size.*

The one-line summary

TorqCapture is **the cheapest new event to stand up after TorqFall** — it fabricates nothing (no gauntlet kit, no castle stock), reuses the entire TorqFall base, and adds only a **zone kit that is almost all electronics.** Flat-taped rings score for ~₹0 of ring hardware; the whole delta is the scoring camera, the controller and — if we want the content-hero lit rings — the LED strip. Budget the electronics honestly and the rest is base you already own.

1. What reuses from the TorqFall base (reference — NOT re-derived here)

Everything in this block is speced + priced in [[TorqFall — What The Event Needs]]; after TorqFall event #1 it is **already owned** and shows up below only as rental-days or per-event consumables. TorqCapture changes **none** of it — it plays on the same turf, uses the same fleet, the same show layer, the same crew and ops kit.

Base block	Reused as-is?	Notes for TorqCapture
Rented fleet + pit (12–16 cars, chargers, pit crew)	Yes — identical	Rulebook §3 reuses TorqFall's 6–8-car heat model; a "match" is just those cars split into Red/Blue. **No siege-ram uplift** (Capture has no heavy attacker). The one add is the strap-on team-colour tag per car — costed in §2 below, not here
Foam bump-rings per car	Yes — identical	Same zip-tied rings; a shove-fest needs them exactly as TorqFall does

Base block	Reused as-is?	Notes for TorqCapture
Turf floor	Yes — **but the 3-ring mode uses the upper end of the roll**	Open play area + runoff; **no 9 m gauntlet lane, no obstacle field.** Single-zone modes (Hill/Hotzone) sit inside the 800 sq ft low end with room to spare, but **3-ring Domination needs the upper ~1,000–1,200 sq ft** of the TorqFall roll — at the tight end of the Rulebook's 4–6 m spacing it is ~1,010 sq ft, and the wide end (6 m centers, 2 m rings ≈ 1,430 sq ft) overruns the roll (fit math in §4 + [[TorqCapture — Full Run Plan]] §2 09:00). Lay Domination at the tight end; single-zone modes have the slack. One-time, owned after TorqFall #1
Soft perimeter barrier (noodle-on-sandbag rail)	Yes — identical	Full-perimeter rail so shoved cars bounce back in, not into the crowd (Rulebook §5). TorqFall stock re-lays around the Capture floor
Drivers' stand	Yes	Red one side, Blue the other; rental day
Show layer (projector/screen + PA + mics + uplights + branded backdrop)	Yes — **works harder**	The big screen is the live scoreboard + flip-colour map (Rulebook §8) — central to Capture, not just dressing. Rental day + the one-time backdrop
Content rig	Yes — **but see §2**	TorqFall's overhead shot was a phone; TorqCapture needs a proper 1080p webcam as the **scoring** camera (it doubles as the content feed). That camera is a delta line in §2, not a re-buy of TorqFall's phone rig
Crew	Yes — + a tech operator	Same 12-ish crew; Capture adds one person to run the CV/scoreboard laptop (Yash or a crew member; ₹1–2k if hired). Marshal is backup-for-disputes, not the scorer
Ops kit + per-event consumables (wristbands, waivers, water, meals, bins, power, transport, insurance)	Yes — identical	Per-event consumables ₹5.5–10k + transport ₹2–5k + insurance ₹10–30k, exactly as TorqFall §8–§9

So the entire TorqCapture-specific spend is §2–§3 below: the zone kit. Everything above is either owned (one-time base) or a standard rental-day/per-event line already derived in the TorqFall doc.

2. The zone kit — scoring rig + tags + lit rings (the delta, from the Tech BOM)

This is the new money. Numbers are lifted directly from [[TorqCapture — Zone-Capture Tech]]'s costed BOM (CV-primary build; the bench test picks CV or the UHF fallback before a rupee goes out — see §6). The BOM splits cleanly into **scoring hardware** (what reads the zones — required) and **LED rings** (the content-hero upgrade — optional; flat-taped or cam-defined rings score for ~₹0).

Scoring hardware — CV primary (required to score):

Item	Qty — why this qty	How we use it	Where	Est ₹
Overhead camera, 1080p USB	1 — one cam covers the **~800–1,200 sq ft turfed play area** (the rings sit inside it, not the full 3000 sq ft spectator floor — §4 turf line) and reads every zone at once; also THE content feed (the flip-colour map, Rulebook §8). **Conditional:** the [[TorqCapture — Zone-Capture Tech]] bench test (its "What to tune" knobs — CV camera height + ring geometry — under *The bench test*) must confirm the truss height + lighting resolve every marker over that footprint; a materially larger play area could need a second stitched cam (Tech *Option C* cons: "one camera covers one floor ... a very large floor might need two stitched views")	Mounted on the content truss looking down; OpenCV finds each car's marker, reads its (x,y), asks "inside ring A/B/C?" → the live per-team count	Logitech C920-class (verified ₹5,150–8,500) — a content buy we'd part-make anyway, but new vs TorqFall's phone-overhead	5,150–8,500
Team-colour marker tops (tags)	**16–20 — one per rented fleet car (12–16) + ~4 spares** (NOT one per live car — every car in the charging rotation keeps its own strap-on marker so it's ready the instant it rolls on; no mid-reset marker-swap — Tech doc fleet-reconcile note)	The strap-mounted ArUco/coloured disc the cam counts; marker colour/ID = team. Velcro/zip-tie on a light foam/card top, faces up, visible all round — nothing bolted (rented fleet)	We print on card/foam + velcro; ₹20–50 each so qty barely moves cost	400–1,000 (lot)
Show laptop (compute)	1 — runs OpenCV + game logic + scoreboard	The brain: count → majority → hold-timer → capture → flip → LED colour + score to the big screen	Already in the shared kit	0
OpenCV / ArUco script	1 — marker-detect → ring geometry → count → capture logic (~200 lines)	Free, open-source; we write it. Mode presets (zone map, hold-timer, relocation interval, points cap) load per mode	We write	0
Cabling, USB extension, overhead mount, power	lot — camera on the truss	Reaches the cam over the floor centre + powers the camera/mount; scoring runs on the free show laptop (no paid controller here)	Hardware store	1,000–2,000

Item	Qty — why this qty	How we use it	Where	Est ₹
CV scoring subtotal (flat-taped rings — no LED)		One camera scores 1–3+ zones; scales cheap. **Excludes the LED-driver ESP32** — it only pushes colour to LED strips, so it isn't needed to score a flat-taped ring; it now lives in the LED block below and re-enters cost only with the LED upgrade		**≈ 6,550–11,500**

Two camera contingencies the bench test decides — NOT in the required subtotal above (named here so nothing un-budgeted is load-bearing):

Optional camera line	When it's bought	Est ₹	Notes
Second overhead camera + stitching code (Domination full-span contingency)	Only if the [[TorqCapture — Zone-Capture Tech]] **full-span coverage trial** (its bench-test trial 6) shows **one** 1080p cam can't resolve markers across the whole ~14.5 m 3-ring Domination axis at a truss height the hall can give	**+ 5,150–8,500** (a 2nd C920-class body; stitching/multi-cam merge is a materially different OpenCV build we write, ₹0)	The Domination A–C span needs ~9–10 m truss for one ~70–78° FOV cam, and a ~10 cm ArUco reads reliably only to ~3 m at 1080p while the far ring is ~11 m slant (Tech doc, Option C cons). If one cam can't span it, the branches are: buy this 2nd cam (each covers ~half the span at a marker-resolving height), OR run Domination as **fewer/bigger rings or marshal-scored** (₹0 — single-zone Hill/Hotzone stay single-cam CV). Decided at the bench test, **before** the T-13 camera buy — not at the T-3 rehearsal
Spare / backup overhead camera (hardware-SPOF insurance)	Optional-recommended; the scoring cam is also Cam 1, a single point of failure	**+ 5,150–8,500** (identical C920-class body)	The **₹0 fallback is already covered** — a phone on the truss running the same ArUco detect is the in-budget stand-in if the cam dies ([[TorqCapture — Full Run Plan]] failure playbook), so this spare is an **explicitly optional buy**, not a required line. Buy it if budget allows so a dead cam is a like-for-like swap, not a phone downgrade. **Not rolled into the required totals below** (same treatment as the Courier cradle)

Team-tag note (derivation): tags = **fleet size, not heat size.** Only 6–8 cars are live per match, but the rented fleet is 12–16 (live + charging rotation + spares, TorqFall §1). A charged car is swapped in every ~30–60 s reset (Rulebook §6, §10), so **every car in the rotation carries its own strap-on marker** — 12–16 + ~4 spares = 16–20. At ₹20–50 each the qty is a rounding error, and it is **already inside the CV scoring subtotal above** (the marker line) — do not add it twice.

LED rings — optional content-hero (arena option 3, Rulebook §5), priced from ring geometry:

A ring is ~1.5–2 m **diameter**, so each ring's circumference is π × 1.5 to π × 2 ≈ **4.7–6.3 m of WS2812B per ring** — more than one 5 m strip at the 2 m end. Cost scales with how many rings light:

LED line	Qty — why	How we use it	Est ₹
WS2812B strips — single-zone modes (Hotzone / The Hill / Siege Point / Courier home zones)	1 ring ≈ 4.7–6.3 m = **1–2 × 5 m strips**	One lit ring, reused across every single-zone mode; changes colour on capture = the content hero	2,400–5,200
WS2812B strips — Domination (3 rings A/B/C)	3 rings × 4.7–6.3 m ≈ **14–19 m = 3–4 × 5 m strips** (a single 5 m strip barely closes ONE 1.5 m ring — three rings need 3–4 strips, not one)	Lights all three Domination zones so the flip-colour map reads on the overhead cam and big screen	7,500–10,400
LED-driver ESP32 (**moved here from the scoring subtotal** — only needed once LEDs are built)	1–2 — only if the laptop doesn't drive the strips directly	Takes the per-zone owner/state from the laptop and pushes WS2812B colour — the LED block's **controller**	250–1,200
5V PSU + power injection (14–19 m Domination run)	1 set — a real 5V supply + injection taps	60 LED/m addressable WS2812B pulls **tens of amps at brightness** over 14–19 m — needs a dedicated 5V 30–40 A supply + injection every few metres (the scoring cabling line is scoped to the camera/USB/mount, not this). The LED block's **power**	1,500–4,000
LED-rings block (Domination) subtotal	strips + LED-driver ESP32 + 5V PSU	Everything the lit 3-ring build needs beyond scoring	**≈ 9,250–15,600**

The Domination 3-ring block (≈₹9,250–15,600 = strips ₹7,500–10,400 + LED-driver ESP32 ₹250–1,200 + 5V PSU/injection ₹1,500–4,000) covers the whole playlist — single-zone modes just light one of the three rings on the same ESP32 + PSU, so we buy the block once, not per mode. The ESP32 and PSU are the LED block's **control + power** — they enter only with the LEDs, never in the flat-taped scoring cost. **To zero the PSU line:** run the rings low-brightness / single-colour so current stays inside a smaller existing supply. **LED is the content upgrade, not required to score:** drop to flat-taped rings (arena option 1) and the ring hardware is ~₹0 (no strips, no ESP32, no PSU) — the CV cam defines each ring in software regardless.

3. Mode-specific bits (small, mostly reused)

The playlist runs five modes on the same rig; only two need any dedicated hardware, and it's cheap.

Item	Qty — why this qty	How we use it	Where	Est ₹
Coloured gaffer — taped rings + Hotzone pre-taped spots	2–4 rolls — mark 3 Domination rings + a handful of pre-taped Hotzone relocation spots across the floor (the controller lights the next spot; no crew on the floor mid-match, Rulebook §4b)	Defines every ring the cam reads; the pre-taped spots let Hotzone "move" in software with zero floor crew	Hardware store	500–1,000

Item	Qty — why this qty	How we use it	Where	Est ₹
Courier neutral core — LED glow orb + spare	2 — 1 live core + 1 spare (Courier is one between-blocks exhibition, Rulebook §4e)	The lit neutral object a car nudges to its home zone to score a delivery; core respawns at centre	Amazon (₹300–600/pc, verified — same orb as TorqSiege)	600–1,200
Courier core cradle (optional, if push-carry needs it)	1 — a foam/printed scoop so a car can shepherd the core cleanly; else cars just nudge it	Holds/pushes the core without bolting anything to a rented car (decided at build)	We print/make	0–300
Raised foam ring lip — **optional**, bench-test decides (arena option 2)	~9–12 noodles if we raise all 3 Domination rings (3 rings × 4.7–6.3 m ≈ 14–19 m ÷ ~1.6 m/noodle) — fewer for one single-zone ring	A low pool-noodle lip so a car nudged to the ring edge rebounds back in and holds position better; tested for whether it traps cars	Wholesale noodles ₹120–250/pc + zip ties (dedicated top-up — TorqFall's rail stock is fully allocated to its own kit)	0–3,000

Siege Point + The Hill need zero dedicated hardware — the lock (Siege Point) and the single central ring (The Hill) are software + a reused ring. The raised-foam lip is a bench-test question (does it help the "hold" feel without trapping cars, Zone-Capture Tech bench test) — priced as an option, ₹0 if flat-taped rings win.

4. The Capture arena (mostly TorqFall's floor, re-marked)

Same turf + barrier stock, a different (and simpler) layout — no gauntlet, just rings. Reconfigure, don't re-buy:

Element	Spec	Built from
3 Domination zones (A/B/C)	~1.5–2 m rings; A and C near each team's end, B central; ~4–6 m apart (far enough you can't hold two, close enough to scramble); every ring ≥2.5 m off the crowd barrier. **Fit math:** the A–C axis spans 2.5 m standoff + 0.75 m radius + two center gaps + 0.75 m radius + 2.5 m standoff in a ~6.5–7 m band — **tight end (1.5 m rings, ~4 m centers) ≈ 14.5 × 6.5 m ≈ 1,010 sq ft (fits the roll); wide end (2 m rings, 6 m centers) ≈ 19 × 7 m ≈ 1,430 sq ft (overruns it)** — so Domination lays at the tight end of the range	Coloured gaffer (§3) ± LED rings (§2) ± raised foam lip (§3); the CV cam defines them in software
Single-zone spots (Hotzone / Hill / Siege Point / Courier)	Pre-taped ring positions across the same floor; the controller lights the live one	Same tape/LED ring, relocated in software

Element	Spec	Built from
Play area turfed + runoff	The play zone + a runoff margin; **not all 3000 sq ft** (spectator floor stays bare)	TorqFall's 800–1,200 sq ft roll — single-zone modes (one ring) sit in the low end with room to spare, but **3-ring Domination needs the upper ~1,000–1,200 sq ft** (fit math above); lay it at the tight end of the 4–6 m spacing, take the 2.5 m standoffs as the runoff margin, and let any overrun spill onto the bare spectator floor rather than buying more turf (no 9 m lane to feed)
Soft perimeter barrier	Full-perimeter noodle-on-sandbag rail, spectators ≥2.5 m off any ring	TorqFall's rail stock, re-laid around the Capture floor
Drivers' stand + big screen	Stand beside the floor (Red / Blue sides); screen = live scoreboard + flip-colour map	Stand rental (shared base) + projector/screen (shared base)
Overhead scoring cam	On the content truss, centred over the **~800–1,200 sq ft turfed play area**, sees all zones inside it (not the full 3000 sq ft floor); height/lighting is a bench-test conditional and the full 3-ring Domination span could need a second stitched cam ([[TorqCapture — Zone-Capture Tech]] *Option C* cons + full-span trial 6; contingency costed in §2)	The C920-class camera (§2) — scores AND shoots

Turf-fit vs the tech branch (the constraint the bench test can force onto the layout):

- **CV wins (expected):** Domination lays at the **tight end** (1.5 m rings, ~4 m centers ≈ 1,010 sq ft) — the CV cam confines in software with no spacing floor, so the tight lay fits the owned roll. This is the default.
- **UHF fallback wins:** UHF's field-spill fix needs **zones ≥6 m apart** ([[TorqCapture — Zone-Capture Tech]] CONDITIONAL). That forces the **wide layout (2 m rings, 6 m centers ≈ 1,430 sq ft), which OVERRUNS the owned 1,000–1,200 sq ft roll** — the tight-lay option that saves turf is **not available under UHF.** So a UHF Domination day needs one of: (a) an **added turf buy** (carry it in contingency — ~₹3–8k for the extra ~230–430 sq ft at the TorqFall roll's per-sq-ft rate), OR (b) a **reduced-ring Domination** (2 zones instead of 3, or the 3 rings partly on **bare concrete** with the ≥2.5 m crowd standoff re-checked so the wider footprint doesn't eat the spectator gap). The generic "overrun spills onto the bare spectator floor" line above is the CV case; the UHF case is spelled out here because under UHF the cheap tight-lay escape is gone. This branch is decided at the bench test and baked into the arena spec then, not discovered on the day.

The number (the Capture delta)

Block	₹
One-time zone kit — CV scoring hardware (camera + markers + cabling) + Courier core + zone tape	≈ 7,650–13,700
— LED rings block (optional content-hero: Domination 3-ring strips + LED-driver ESP32 + 5V PSU/injection, reused across all modes)	+ 9,250–15,600

Block	₹
— Raised foam ring lip (optional, bench-test decides)	+ 0–3,000
One-time zone kit — flat-taped rings (scoring only)	**≈ 7,650–13,700**
One-time zone kit — full (lit rings + foam lip)	**≈ 16,900–32,300**
Recurring per event	≈ 500–1,500 (a broken marker, a tape re-buy, an orb battery — the electronics + markers reuse every event, strap-on/comes off clean)

Arithmetic (rows reconcile to the totals): flat-taped = CV scoring subtotal (§2, ₹6,550–11,500 — camera + markers + cabling; the LED-driver ESP32 has moved to the LED block, so it isn't in the scoring-only cost) + Courier core, 2 orbs (§3, ₹600–1,200) + zone tape/coloured gaffer (§3, ₹500–1,000) = **₹7,650–13,700**. Full = flat-taped (₹7,650–13,700) + LED-rings block (§2, ₹9,250–15,600 = Domination 3-ring strips ₹7,500–10,400 + LED-driver ESP32 ₹250–1,200 + 5V PSU/injection ₹1,500–4,000) + foam lip (₹0–3,000) = **₹16,900–32,300**. The optional Courier cradle (§3, ₹0–300) is a build-time call and is **not** rolled into these totals; add it only if push-carry needs it.

On top of the shared TorqFall base (cars, floor, show, stand, crew, ops — of which turf, barrier, ops kit and content rig are already owned after TorqFall event #1). This is the **smallest delta of any game in the set**: TorqFall adds a whole gauntlet kit (₹40–61k), TorqSiege adds castle stock + weapons (₹22.5–42.5k one-time + ₹3–12k/event consumables), TorqCapture adds **just electronics** (₹7.65–32.3k one-time — ₹7.65–13.7k for the flat-taped scoring build, up to ₹32.3k only if the full lit rings + foam lip are built — plus ~₹0.5–1.5k/event) because it fabricates nothing and consumes nothing. The flat-taped scoring build is by far the smallest one-time spend in the set, and even the full lit build stays under TorqFall's and TorqSiege's one-time ceilings with near-zero per-event cost.

The full event number (standalone TorqCapture day — the ₹4L envelope check)

TorqCapture is its own standalone event, so the delta alone isn't a budget. Assumes TorqFall event #1 has run (locked: TorqFall is event #1), so the shared one-time base — turf, barrier, ops kit, content rig, branding — is already owned and shows up only as rental-days / consumables / re-lay below. Low end = flat-taped rings + projector + friends-on-cams route; high end = lit rings + foam lip + everything hired + top insurance quote.

Block	₹
Cars + pit (12–16 rented, per TorqFall §1 — no siege-ram uplift)	17,000–32,000
Capture one-time zone kit (first Capture event only — flat-taped low end → full lit+foam high end)	7,650–32,300
Floor re-lay (tape; turf + barrier owned)	3,000
Drivers' stand (rental day)	4,000–8,000
Show layer (rental day, projector route — the screen is the live scoreboard)	10,000–20,000

Block	₹
Hired crew (fully hired, per TorqFall §7) + scoreboard/tech operator ₹1–2k if not a friend	14,000–27,000
Ops per-event consumables (wristbands, waivers, water, crew meals, bags)	5,500–10,000
Podium — trophy + team medals (a team game crowns a **team** of 5–6, so a medal set, not one cup — the run plan's 17:45 podium)	500–3,000
Transport (tempo/van)	2,000–5,000
Public-liability insurance (per event — Event Ops §5 must)	10,000–30,000
Subtotal	≈ 0.74–1.70 L
Contingency ~10%	7,400–17,000
Total ex-venue	**≈ 0.8–1.8 L**

Envelope check: ₹4L all-in − ex-venue ₹0.8–1.8L = **₹2.2–3.2L left for the venue** — closes with the most headroom of any game in the set, because the shared base is already owned AND the Capture delta is the smallest (electronics only). Even the full lit-rings, everything-hired, top-insurance build leaves ~₹2.2L for the room. *(The ₹500–3,000 podium line firms the high-end ex-venue to ~₹1.87L including contingency — inside the rounded ≈ ₹1.8L band, exactly as the pre-podium high end was ~₹1.84L; it does not move the rounded ex-venue or the ₹2.2–3.2L headroom.)*

From Capture event #2 the one-time zone kit drops out (camera, controller, markers, LED all reused — strap-on, comes off clean) and only the ~₹0.5–1.5k recurring line enters: ex-venue ≈ **₹0.7–1.5L** — the same shape as TorqFall event #2, since Capture adds essentially no per-event consumable. *(The backup-power decision ₹0–5k and the ₹0 rehearsal line apply to any standalone event per [[TorqFall — What The Event Needs]] §9 — they sit inside contingency here.)* **If TorqCapture ever ran before any TorqFall event** — it won't, TorqFall is locked as #1 — add the shared one-time base it would otherwise inherit: floor ₹27–69k + barrier/ops kit ₹20–30k + content rig ₹4–11k + branding ₹3–8k (per the TorqFall doc's numbers). No gauntlet kit is ever needed — Capture doesn't use it.

Before spending anything (the gate — same spirit as TorqSiege's collapse test)

Run the **bench test** in [[TorqCapture — Zone-Capture Tech]] first: 2–4 borrowed cars, one taped 1.5–2 m ring, the CV rig (phone/webcam overhead + OpenCV) and the UHF rig side by side on the same ring, an afternoon, near-zero cost. It answers the one gating question — **can the reader count the cars per team inside the ring, and only that ring, live, while they shove and overlap?** — and picks the tech (expected: CV) + freezes the ring size + hold-timer + whether the raised foam lip helps.

- **PASS** → lock CV, buy the §2–§3 list above, build the event.
- **CONDITIONAL** (e.g. CV needs a specific cam height/lighting, or the fallback needs zones ≥ 6 m apart) → buildable, bake the constraint into the arena spec + run plan.

- **FAIL** (neither CV nor UHF holds a boundary + count under a pile-up) → fall back to a **marshal-scored** version (bigger/fewer zones, a human calling majority — slower, less content, but the game still runs) while the tech is reworked. Don't commit the full electronics build until the bench test passes.

The only lines worth buying pre-test: a webcam + printed markers + a tape ring (the test itself needs them). If the fallback ends up winning, it's a **swap, not an add** — the CV camera + printed CV markers come out, **3 power-tuned UHF readers + antennas + passive tags ≈ ₹14–22k** go in ([[TorqCapture — Zone-Capture Tech]] *UHF fallback* table). The free show laptop and the scoring cabling are shared, but UHF needs a **paid ESP32 reader-aggregator + cabling ≈ ₹1.25–3.2k** for a like-for-like scoring build ≈ **₹15.25–25.2k**. Add the same Courier core + zone tape (₹1.1–2.2k) and the **UHF one-time zone kit lands at ≈ ₹16–28k** (₹15,250–25,200 + ₹600–1,200 + ₹500–1,000 = ₹16,350–27,400) — **not** the ₹22–37k a camera-*kept* build would cost, because the camera is swapped out, not retained (matches the Tech doc's "swaps the camera for per-zone readers"). Still the cheapest new event in the set. This UHF kit is the flat-taped (no-LED) comparator to the CV flat-taped ₹7.65–13.7k; LED rings, if built, add the same ₹9.25–15.6k block on top of either tech.

Start-today order once the bench test passes: (1) confirm the winning tech + freeze ring size/hold-timer, (2) buy the camera + print markers + wire the controller, (3) write/finish the OpenCV script + mode presets, (4) build LED rings only if the content-hero look is wanted, (5) everything else is the TorqFall base you already own.

Doc-set status (what this doc is and isn't)

This is the **kit + budget** doc — it costs the event, it does not run the day. The end-to-end operational play (pre-event timeline, build-morning checklist, hour-by-hour run-of-show, failure playbook, post-event content/debrief loop) is the job of `TorqCapture — Full Run Plan.md`, which both sibling standalone events ship. That doc is **now built and on disk** (its own 2026-09-06 changelog) — so the TorqCapture set holds **five docs**: the Design Study, Rulebook, Zone-Capture Tech, this doc, and the Full Run Plan.

- **Run-of-show now has two layers, both present:** the Full Run Plan is the primary end-to-end ops source; [[TorqCapture — Rulebook]] **§6** (the match script — preset load → tag check → "3-2-1-GO" → live scoring → horn → 30–60 s reset) and **§10** (reset math) remain the quick-reference backup for the match loop, and historically **housed the interim run-of-show while the Full Run Plan was still in-build.** The bench-test gate above and [[TorqCapture — Zone-Capture Tech]] carry the pre-spend sequence. This is kept in sync with the TorqCapture Notion Game Page (which now lists the Full Run Plan as a present, built file).

Price sources (Sept 2026): *all electronics prices pulled from [[TorqCapture — Zone-Capture Tech]]'s verified BOM — Logitech C920-class 1080p webcam ₹5,150–8,500 (IndiaMART/Computech/Moglix/LT Online); ESP32 dev boards ₹250–600 (Robokits/Robu/Zbotic); WS2812B 5 m addressable strip ₹2,400–2,600 (Flipkart mdblck ₹2,499, Robu, Amazon); passive UHF tags ₹15–25 + YRM100-class readers ~₹4,500–7,000 each (Alibaba/AliExpress import) for the fallback; ArUco markers + OpenCV = free (we print/write). LED glow orb ₹300–600 (Amazon.in, same as TorqSiege). Pool noodles wholesale ₹120–250/pc, coloured gaffer + zip ties (hardware store) — per the TorqFall doc's verified sources. All base-block prices (cars, turf, stand, show, crew, ops, transport, insurance) are the TorqFall/TorqSiege verified numbers, not re-quoted here.*

Verified + changelog (2026-09-06): *first build, in the TorqSiege "delta + full standalone number + ₹4L envelope check" format. Structured as (1) what reuses from the TorqFall base — reference table, not re-derived; (2) the zone kit delta with every electronics number pulled directly from the Zone-Capture Tech BOM (CV scoring subtotal ≈ ₹6.8–12.7k incl. the overhead camera + markers + controller + cabling; LED rings split as the optional content-hero, Domination 3-ring set ₹7.5–10.4k reused across all modes; UHF fallback flagged); (3) mode-specific bits (coloured gaffer for taped/Hotzone-relocation rings, 2 Courier orbs, optional raised foam lip priced from ring geometry); (4) the arena (TorqFall's turf re-marked, no gauntlet). Quantities derived line-by-line: tags = fleet size (16–20, one per rotation car, not heat size — matches the Tech doc's fleet-reconcile note, and stated as already-inside the CV subtotal so it isn't double-counted); Domination = 3 rings, single-zone modes reuse one; noodle count for the foam lip from ring circumference ÷ noodle length. Delta ≈ ₹7.9–14.9k flat-taped / ₹15.4–28.3k full (lit+foam), ~₹0.5–1.5k/event recurring — stated as the smallest delta of any game (fabricates + consumes nothing; electronics are the one genuinely new spend). Full standalone event ex-venue ≈ ₹0.8–1.8L, venue headroom ₹2.2–3.2L (the most of any game — owned base + smallest delta); event #2+ ex-venue ≈ ₹0.7–1.5L. Bench-test gate stated as the pre-spend condition (same role the collapse test plays for TorqSiege), with the UHF-fallback delta shift (₹22–37k one-time) noted honestly. No siege-ram uplift, no gauntlet kit — both explicitly excluded.*

Amendment (2026-09-06, cost re-derivation pass): *reconciled "The number" table to its own §2–§3 line items after an examiner flagged three totals that didn't add up. (1) Flat-taped (scoring-only) corrected ₹8,100–15,100 →* ***₹7,900–14,900*** *= CV subtotal ₹6,800–12,700 + Courier core ₹600–1,200 + zone tape ₹500–1,000; the ₹0–300 Courier cradle is explicitly excluded from the total (build-time call). (2) Full (lit+foam) corrected ₹16,600–28,500 →* ***₹15,400–28,300*** *= flat-taped + LED ₹7,500–10,400 + foam lip ₹0–3,000; added an explicit arithmetic line so rows and totals reconcile. Knock-ons: full-event zone-kit line 8,100–28,500 → 7,900–28,300; rounded delta callout ₹8–28.5k → ₹7.9–28.3k; subtotal-low shift is ~₹200 (₹73.6k → ₹73.4k), inside rounding, so ex-venue stays ≈ ₹0.8–1.8L. (3) Camera-coverage claim corrected: the overhead cam covers the* ***~800–1,200 sq ft turfed play area*** *(where the rings sit), not the full 3000 sq ft spectator floor, with the height/lighting bench-test conditional and possible second stitched cam cross-referenced to [[TorqCapture — Zone-Capture Tech]] §57–§58 — in §2 and the §4 arena table.*

Amendment (2026-09-06, costs-stream pass): *fixed four examiner findings, mirroring the BOM changes in [[TorqCapture — Zone-Capture Tech]]. (1)* ***UHF fallback was overstated*** *— the old "₹22–37k" reachable only by keeping the ₹5.15–8.5k camera AND adding UHF readers, which contradicted the "delta shifts from the camera to readers" wording and the Tech doc's "swaps the camera." Rewritten as a true* ***swap*** *(camera + CV markers out, readers + tags in): UHF like-for-like scoring ≈ ₹15.25–25.2k + Courier core + zone tape ⇒* ***UHF one-time zone kit ≈ ₹16–28k*** *(₹16,350–27,400), with explicit arithmetic, and stated NOT to be the camera-kept ₹22–37k. (2)* ***Mis-bucketed LED-driver ESP32*** *— it sat in the flat-taped CV scoring subtotal but only drives LED strips; moved into the LED block, so scoring-only subtotal drops* ***₹6,800–12,700 → ₹6,550–11,500*** *and the ESP32 re-enters only with the LED upgrade. (3)* ***Broken §57–§58 citations*** *— the Tech doc uses named sections; replaced with named targets (_The bench test* "What to tune" knobs; *Option C* overhead-cam cons "a very large floor might need two stitched views") in §2 and §4. (4) **LED block missing power** — added a **5V PSU + power-injection** line (₹1,500–4,000 for 14–19 m of WS2812B; zero-able by running rings low-brightness/single-colour), with the moved ESP32 the LED block's control and the PSU its power. **Cascaded totals:** flat-taped ₹7,900–14,900 → **₹7,650–13,700** (CV subtotal 6,550–11,500 + Courier core 600–1,200 + zone tape 500–1,000); LED-rings block now strips+ESP32+PSU ≈ **₹9,250–15,600**; full (lit+foam) ₹15,400–28,300 → **₹16,900–32,300**; full-event zone-kit line 7,900–28,300 → 7,650–32,300;

delta callout ₹7.9–28.3k → ₹7.65–32.3k; full-event subtotal high 1.64L → 1.67L, contingency high 16k → 17k — **ex-venue stays ≈ ₹0.8–1.8L and venue headroom ≈ ₹2.2–3.2L unchanged** (high ex-venue ₹1.84L ⇒ ₹2.16L for the room). Arithmetic line and "The number" table re-reconciled to the new §2–§3 rows._

Amendment (2026-09-06, notion-stream pass): *added a **"Doc-set status"** section after an examiner flagged that this doc — like both sibling standalone events — sits alongside a not-yet-built `TorqCapture — Full Run Plan.md`, with no in-doc pointer to where the run-of-show currently lives. Added the explicit in-build disclosure and the interim-run-of-show cross-reference to [[TorqCapture — Rulebook]] §6/§10 (matching the Notion Game Page and Design Study disclosures), so the deferral is verifiable as deliberate. No cost numbers touched.*

Amendment (2026-09-06, notion-stream sync pass): *the `TorqCapture — Full Run Plan.md` is now built and on disk (its own 2026-09-06 changelog), so the "Doc-set status" section was corrected — the earlier "in-build (not yet on disk) … set currently holds four docs" claim is stale. Revised to state the set now holds **five docs** and to reframe the Rulebook §6/§10 as the quick-reference backup (which historically housed the interim run-of-show), kept in sync with the Notion Game Page. No cost numbers touched.*

Amendment (2026-09-06, run-plan stream reconcile): *the run-plan examiner caught that the **turf "room to spare" claim contradicts the 3-ring Domination footprint.** Added the A–C **fit math** to §4 (tight end 1.5 m rings / ~4 m centers ≈ 1,010 sq ft fits the roll; wide end 2 m rings / 6 m centers ≈ 1,430 sq ft overruns it) and corrected the "room to spare" wording in §1 (Turf floor row) and §4 (Play area row): single-zone modes keep the low-end slack, but **3-ring Domination needs the upper ~1,000–1,200 sq ft and lays at the tight end of the Rulebook's 4–6 m spacing**, with the 2.5 m standoffs as the runoff margin and any overrun spilling onto the bare spectator floor. Matches [[TorqCapture — Full Run Plan]] §2 (setup 08:00/09:00). No cost numbers touched.*

Amendment (2026-09-06, run-plan stream pass 2): *fixed three more examiner findings (kit/cost side of the run-plan defects). (1) **CV full-span (Domination) contingency named + costed** — §2 now carries a **second overhead camera + stitching-code** contingency (+₹5,150–8,500, decided by the Tech doc's full-span bench trial 6 **before** the T-13 camera buy) with the alternative of **fewer/bigger rings or marshal-scored Domination** (₹0); mirrors the new coverage math in [[TorqCapture — Zone-Capture Tech]] Option C. Not in the required subtotal (a bench-test-gated contingency, like the cradle). (2) **Spare scoring camera named as an explicitly optional buy** — §2 lists a spare/backup overhead camera (+₹5,150–8,500) as **optional-recommended**, with the **₹0 phone-on-truss** stated as the in-budget SPOF fallback, so the run-plan's camera-failure defense no longer leans on an un-budgeted item; **not rolled into the required totals**. (3) **UHF-fallback ≥ 6 m spacing vs turf-fit reconciled** — §4 now spells out that a UHF Domination day is forced to the wide (~1,430 sq ft) layout that **overruns the owned roll** (the tight-lay CV escape is unavailable under UHF), so it needs either an **added turf buy** (~₹3–8k, carried in contingency) or a **reduced-ring/2-zone or bare-concrete Domination** with the ≥ 2.5 m standoff re-checked. (4) **Podium costed** — added a **trophy + team medals** line (₹500–3,000/event) to the full-event table, since a team game crowns a team of 5–6; subtotal 0.73–1.67L → **0.74–1.70L**, contingency 7,000–17,000 → 7,400–17,000, high-end ex-venue firms to ~₹1.87L (inside the rounded ≈ ₹1.8L band, as the pre-podium high was ~₹1.84L) — **ex-venue ≈ ₹0.8–1.8L and headroom ₹2.2–3.2L unchanged**. Mirrors [[TorqCapture — Full Run Plan]] fixes.*